Point-by-Point Response to Reviewers

Manuscript PR-D-26-04476

Interpretable supervised feature extraction for time series and spatial data

June 6, 2026

Dear Editor, Associate Editor, and Reviewers,

We thank you for the constructive major-revision decision and for the detailed feedback, which has substantially improved the manuscript. Below we respond to every point. We address **Reviewer #2**'s suggestions through the editorial questionnaire (Q1–Q8) and answer **Reviewer #1**'s metareview point by point. For each item, the lead sentence summarizes what we changed; bullets quote key wording from the **revised** manuscript (in quotation marks).

The revised manuscript complies with the journal's **35-page limit**. Headline changes include: numbered Objectives and a condensed Related Work with expanded Table 1; Methods renamed to *Iterative Feature Discovery* (fit/selection halves); baseline hyperparameters and dataset Table 2 in the main text; separate Supplementary Data (preprocessing only); paired Wilcoxon tests and per-patient AZT1D reporting; multi-fold ablations with a *Resampling* variant; expanded REMC/AZT1D SHAP interpretation; and a general reframing regarding algorithm superiority and clinical applicability. Several requests could not be implemented at full depth in the main text without exceeding that limit (e.g. step-by-step preprocessing for all datasets, multi-seed repeats of the full feature search, 95% confidence-interval columns, additional spatial benchmarks, SHAP on every dataset, or high-resolution supplementary figure files). Where we defer detail, we say so explicitly and provide the closest alternative—usually the separate Supplementary Data document or a condensed main-text summary.

Reviewer comment

1. Are the objectives and the rationale of the study clearly stated?

Reviewer #1: *Yes.*

Reviewer #2: *The manuscript presents a clear overall motivation—addressing the trade-off between predictive performance and interpretability in time series feature extraction—and this is reasonably well justified in the introduction; however, the objectives are not explicitly stated and remain implicit throughout the text. To improve clarity, the authors should include a concise and explicit statement of the main objective at the end of the introduction, more clearly separate the problem motivation from the methodological contribution, and articulate specific research goals or guiding questions. In addition, the novelty of the approach should be more sharply defined relative to existing methods (e.g., interval-based or shapelet approaches), and the stated objectives should be better aligned with the evaluation, particularly with respect to how interpretability is assessed.*

Our response

We now state the study objectives explicitly. The revision adds Sec. 1.1 (Objectives) with three aims and a condensed Sec. 1.2 (Related Work) before Methods.

- Sec. 1.1 (new): "The objectives of our study are as follows: (i) we present SublimeX ... (ii) we benchmark SublimeX against other existing methods on seven public biomedical datasets, and (iii) we assess robustness, generalization and interpretability through post-hoc analyses and ablations."
- Sec. 1.2 (revised): "SublimeX combines traceable segment means with supervised search over channel, transform, and location"; Table 1 compares methods by type, approximate feature count, multivariate handling (Yes = joint across channels; No = per-channel concatenation), and interpretability (High/Medium/Low with brief rationale).

Reviewer comment

2. If applicable, is the application/theory/method/study reported in sufficient detail to allow for its replicability and/or reproducibility?

Reviewer #1: *Yes [X].*

Reviewer #2: *The manuscript provides a generally solid methodological description of the proposed framework (including the optimization procedure, transformations, and evaluation setup), which supports partial reproducibility; however, several details remain insufficient for full replication. To improve reproducibility, the authors should (1) specify all implementation details more precisely, including exact hyperparameters of the downstream models (beyond default mentions) and stopping criteria, (2) clarify how segment boundaries are sampled and constrained during optimization, (3) provide full details of preprocessing steps for each dataset within the main text rather than relying on supplementary material, (4) describe the exact data splits or make them publicly available to ensure consistent benchmarking, (5) report random seeds and clarify whether results are sensitive to initialization, (6) include computational cost details such as total runtime per dataset and hardware variability, and (7) provide direct access to code, configuration files, and scripts used to reproduce experiments, ideally with instructions for end-to-end replication.*

Our response

We expanded Methods so the study can be reproduced end to end. Because of the 35-page limit, we kept the per-dataset preprocessing to a **separate Supplementary Data** file. However, we state all Baseline parameters clearly in the main text.

- Methods, Iterative Feature Discovery (renamed from "Segment-Based Feature Optimization"): "center and width on a normalized 0–1 scale"; a fixed 50/50 split of the training data into a "*fit half*, used to train the downstream model, and a *selection half*, used to score candidate features"; "terminates once no candidate yields a strict improvement"; fit/selection terminology replaces the earlier search-train/search-validation wording.
- Methods, Baselines: tsfresh, catch22, MiniRocket, RDST, and CNN hyperparameters in prose (replacing the former supplementary "Baseline Implementations" appendix).
- Methods, Implementation Details: "300 trials per proposed feature"; "maximum tree depth 5"; "published as an open-source Python package on PyPI (version 0.1.2)"; "MacBook Pro M2 (12 cores, 32 GB RAM)."
- Table 3: now reports performance, feature counts, and wall-clock time.
- Manuscript repository: `https://github.com/Prgrmmrjns/SublimeX-manuscript` contains step-by-step reproduction instructions (data download, experiment pipeline, tables/figures, LaTeX build)

Reviewer comment

3. If applicable, are statistical analyses, controls, sampling mechanism, and statistical reporting (e.g., P-values, CIs, effect sizes) appropriate and well described?

Reviewer #1: *Yes [X].*

Reviewer #2: *The manuscript includes standard evaluation practices (e.g., cross-validation, reporting mean ± standard deviation across folds) and uses appropriate performance metrics depending on the task (AUC, accuracy, RMSE), which supports a generally acceptable level of statistical reporting; however, the statistical analysis lacks depth and rigor in several aspects. To improve this, the authors should (1) include statistical significance testing when comparing methods (e.g., paired tests across folds or datasets) to support claims of competitive performance, (2) report confidence intervals in addition to standard deviations to better quantify uncertainty, (3) clarify whether hyperparameter tuning and feature selection are nested within cross-validation to avoid potential data leakage, (4) provide more detail on the sampling strategy and class balance handling for each dataset, particularly in imbalanced settings, (5) assess variability across runs by repeating experiments with different random seeds where applicable, and (6) better justify the choice of evaluation protocols (e.g., single split vs. cross-validation) and discuss their potential impact on the reported results.*

Our response

We strengthened statistical reporting with paired Wilcoxon tests, explicit evaluation protocols, and anti-leakage wording. For cross-validated benchmarks, tests pair matched folds; for AZT1D we fully re-ran all methods so each of 23 patients contributes one held-out test RMSE and significance is assessed on that patient-level distribution (not across folds). Table 3 reports AZT1D as mean±SD over those 23 per-patient errors (e.g. SublimeX 30.98 ± 5.65 vs. MiniRocket 30.76 ± 5.38, p=0.39). We do not add separate 95% confidence-interval columns (five folds yield wide intervals, largely redundant with mean±SD) or repeat the full feature search across random seeds.

- Methods, Evaluation: "paired two-sided Wilcoxon tests (α=0.05)" on matched folds, LOSO subjects, or patients; Table 3 asterisks; anti-leakage wording; PAMAP2 once-assigned fit/selection split.
- Methods, Evaluation (AZT1D): temporal 80/20 split per patient; pooled segment discovery on training windows only; RMSE as mean±SD over n=23 per-patient test errors (full re-run for all methods).
- Methods, Iterative Feature Discovery: fit/selection 50/50 split during search; test scoring in Evaluation.
- Table 4: multi-fold ablations with mean±SD (replacing single-fold ablations).

Reviewer comment

4. Could the manuscript benefit from additional tables or figures, or from improving or removing (some of the) existing ones?

Reviewer #1: *No.*

Reviewer #2: *The manuscript includes several relevant figures and tables that support the methodology and results (e.g., workflow diagram, performance comparison, and feature analysis), but their effectiveness could be improved. Specifically, (1) the graphical abstract is visually appealing but somewhat generic and could be made more informative by explicitly illustrating the optimization process, (2) Figure 1 (methodology) is helpful but visually dense and would benefit from simplification or clearer labeling of key steps, (3) Figure 2 and Figure 3 provide useful insights but are difficult*

to interpret due to small text and multiple panels, so readability should be improved (e.g., larger fonts, clearer legends), (4) Table 3 would benefit from including statistical significance indicators or highlighting meaningful differences beyond bolding best values, (5) an additional table summarizing the number and type of extracted features per dataset would strengthen interpretability claims, and (6) a more explicit visualization comparing interpretability (not only performance) across methods would further support the main contribution.

Our response

We improved the graphical abstract and Figures 1–4 for readability and extended Tables 1 and 3 (Wilcoxon markers, runtimes, and new Multivariate and Interpretability columns).

- Graphical abstract: redrawn with larger font size and SublimeX workflow from flowchart
- Figure 1: redrawn flowchart; caption states the loop stops when "evaluation no longer improves."
- Figure 2: regenerated with larger fonts (transform composition and pattern stability).
- Figure 3: incremental train/validation/test curves; AZT1D uses pooled training windows matching the feature-discovery protocol.
- Figure 4: full-width six-panel SHAP layout for REMC and AZT1D, only top 5 features instead of all features.
- Table 3: "wall-clock time" row and significance asterisks.
- Table 1: added Multivariate (Yes/No) and shortened Interpretability columns (High/Medium/Low with brief rationale); method types written out (Segment, Interval, Convolution, DL, etc.).

Reviewer comment

5. Have the authors clearly emphasized the strengths of their study/theory/methods/argument?

Reviewer #1: *Yes.*

Reviewer #2: *The manuscript demonstrates several clear strengths—particularly the balance between interpretability and performance, the modular design of the framework, and the use of diverse datasets—but these strengths are not always emphasized as clearly or strategically as they could be. To improve this, the authors should (1) explicitly summarize the main strengths of the method in a dedicated contributions paragraph at the end of the introduction, (2) more clearly highlight the novelty of combining supervised feature search with interpretable, segment-based representations, (3) emphasize the flexibility and modularity of the framework as a practical advantage for different domains, (4) reinforce the importance of compact feature sets as a key contribution rather than a secondary outcome, and (5) more directly connect the empirical results and SHAP analysis to the claimed strengths in interpretability and real-world applicability.*

Our response

We make the key strengths—modularity and compact, traceable features—explicit in the Objectives and Implementation Details, while keeping the claims balanced against the limitations. The Discussion and Conclusion use measured, "competitive" wording.

- Sec. 1.1 (Objectives): compact traceable features linked to ablations and interpretability analyses.
- Table 1: multivariate Yes/No and interpretability ratings (High/Medium/Low) make the performance–interpretability trade-off explicit across method families.

- Methods, Implementation Details: "modular, extensible design"; customizable objective, model, and transform library.
- Discussion and Conclusion:rephrased "competitive" wording; REMC before AZT1D in Summary; PAMAP2 LOSO interpretation in Discussion.

Reviewer comment

6. Have the authors clearly stated the limitations of their study/theory/methods/argument?

Reviewer #1: *Yes.*

Reviewer #2: *The manuscript acknowledges several important limitations—such as computational cost, potential overfitting during feature search, and challenges in cross-subject generalization—which is a clear strength; however, these limitations could be stated more explicitly and systematically. To improve this, the authors should (1) consolidate the limitations into a clearly structured paragraph or subsection for better visibility, (2) more explicitly discuss the impact of the greedy search strategy and its susceptibility to local optima, (3) elaborate on the risk of overfitting due to the fixed train/validation split used during optimization, (4) clarify the scalability limitations in terms of runtime and high-dimensional search space, (5) more critically address the limited comparison with additional state-of-the-art methods, and (6) discuss the potential limitations of interpretability, particularly whether simple segment-based features are sufficient for capturing complex patterns in all domains.*

Our response

We expanded Discussion, Limitations: greedy search/local optima (with parallel ablation caveat), PAMAP2 validation–test gap (∼26%) and fixed-split risk supported by the *Resampling* ablation, omitted subject-holdout search for compute cost, slower runtime vs. lighter baselines (Table 3), and narrower TSC/UCR comparison scope. Future Directions adds nested internal validation.

- Discussion, Limitations: greedy search; PAMAP2 gap; *Resampling*; runtime; TSC/UCR scope.
- Discussion, Future Directions: nested or repeated internal validation.

Reviewer comment

7. Does the manuscript structure, flow or writing need improving (e.g., the addition of subheadings, shortening of text, reorganization of sections, or moving details from one section to another)?

Reviewer #1: *No.*

Reviewer #2: *The manuscript is generally well organized and follows a standard structure, but the clarity and flow can be improved in several areas. Specifically, (1) the introduction would benefit from a clearer closing paragraph summarizing the main contributions in a structured way, (2) some sections—particularly the methodology—are dense and could be streamlined by reducing repetition and improving transitions between subsections, (3) key implementation details currently placed in supplementary material should be briefly summarized in the main text to improve readability, (4) the results section could be better structured by explicitly linking each subsection to a specific research objective or claim, and (5) the discussion could be made more concise by reducing redundancy and more clearly separating interpretation, limitations, and future work.*

Our response

We reorganized the manuscript for clearer flow within the 35-page limit: explicit Objectives, a condensed Related Work, a separate Supplementary Data file, and a four-part Discussion.

- Sec. 1.1–1.2: explicit objectives and a streamlined related-work contrast before Methods, with Table 1 summarizing method type, feature count, multivariate handling, and interpretability; smoother transitions between method families.
- Methods: "Segment-Based Feature Optimization" renamed *Iterative Feature Discovery*; reduced repetition and improved transitions across Signal Transformation, Iterative Feature Discovery, and Implementation Details.
- Methods, Datasets: Table 2 in main text (short caption; "channels × length"); MIMIC-IV initial-feature block described; preprocessing in Supplementary Data.
- Methods, Baselines: library and CNN hyperparameters in main text (Section Baselines); LightGBM readout in Implementation Details (depth 5).
- Results: Performance Comparison, Transform Composition, Incremental Feature Analysis (PAMAP2 numbers only; LOSO interpretation in Discussion), Ablation Study, Domain-Specific Feature Interpretation as distinct subsections.
- Discussion: Summary, Comparison to Related Work, Limitations, and Future Directions as separate subsections.

Reviewer comment

8. Could the manuscript benefit from language editing?

Reviewer #1: *Yes.*

Reviewer #2: *Yes.*

Our response

We carried out a manuscript-wide language and flow edit: softened overstated wording, standardized dataset names, and improved readability while staying within the 35-page limit (the Abstract was condensed to ≤250 words).

- Introduction: tighter motivation and a clearer contribution lead-in ("To this end we present **SublimeX**…"); Related Work now reads as a single contrast that ends with "SublimeX bridges these strands by pairing traceable segment means with supervised search."
- Methods: smoother Signal Transformation and Iterative Feature Discovery paragraphs and a crisper Implementation Details (modularity) passage, with no change to reported settings.
- Discussion: shorter paragraphs; opens with a SublimeX recap and measured performance language; separate Limitations / Future Directions subsections.
- Conclusion (revised): three generalized sentences on competitive traceable features, interpretability vs. performance, and open-source availability.
- Throughout: consistent dataset naming (MIMIC-IV, MIT-BIH, etc.) in Tables 2–4 and text.

Reviewer comment

Associate Editor (Reviewer #1) — full metareview

The manuscript "Interpretable supervised feature extraction for time series and spatial data", I would recommend major revision. The topic is relevant and the proposed SublimeX framework has potential, but several methodological, validation, and presentation issues need to be addressed before

publication.
The manuscript presents an interesting interpretable feature extraction framework, but the novelty of SublimeX should be stated more clearly. The authors should explain how it differs substantially from existing interval-based, shapelet-based, ROCKET/MiniROCKET, and supervised feature extraction methods. The title mentions both "time series and spatial data," but most of the paper focuses on time-series biomedical datasets. The spatial data contribution is not sufficiently demonstrated. The authors should either strengthen the spatial-data evaluation or revise the title and scope. The abstract is informative, but it overstates competitiveness. SublimeX is not the best-performing method on several datasets, including MITBIH, Emotions, MIMIC-IV, PAMAP2, and SVD. The abstract should more cautiously describe the results. The methodology needs more detail on the stopping criterion. The paper states that the loop stops when no candidate improves performance, but the exact definition of improvement, tolerance threshold, and validation metric direction should be clearly specified. The use of a single deterministic 50/50 train/validation split during feature optimization may introduce instability and overfitting. The authors should justify this choice and discuss whether nested cross-validation or repeated validation would provide more reliable feature selection. The internal feature-search validation split does not match the external evaluation design for some datasets, especially PAMAP2 with leave-one-subject-out validation. This likely explains the large validation-test gap. The authors should consider using subject-independent validation during feature search. The manuscript should clarify whether feature extraction is performed independently within each training fold only. If any information from validation or test folds influences feature discovery, this would create data leakage. The comparison with baselines is incomplete. Important state-of-the-art time series methods such as HIVE-COTEv2, InceptionTime, and LSTM/GRU-based models are only mentioned as limitations but not evaluated. At minimum, the authors should justify why these were excluded. The authors should provide statistical significance testing across methods. Many reported differences are very small, such as AZT1D RMSE 31.11 vs. 31.18 and REMC AUC 0.931 vs. 0.931, so claims of superior performance require statistical support. The results show that SublimeX is competitive but not consistently superior. The discussion should better balance performance and interpretability rather than implying broad performance advantage. The ablation study is based on a single fold only, which limits the reliability of the conclusions. The authors should either repeat ablations across multiple folds or clearly tone down the strength of the claims. Runtime comparisons need clearer reporting. The manuscript reports runtimes for ablation variants, but does not compare computational cost against all baseline methods under the same hardware and settings. The graphical abstract and Figure 1 contain small text and may be difficult to read. The authors should improve figure resolution, font size, and visual clarity. Some figures, especially SHAP visualizations, are too small to interpret in the manuscript. Larger panels or supplementary high-resolution versions should be provided. The paper should provide the exact hyperparameters used for LightGBM, CNN, MiniRocket, RDST, tsfresh, and catch22 to ensure reproducibility. The authors state that SublimeX is open source, but the manuscript should include a permanent repository link, version number, and ideally a reproducibility package for all experiments. The manuscript should better define what "interpretable" means in this context. Traceability to a signal segment is useful, but interpretability also depends on clinical or biological validity, stability, and expert confirmation. The SHAP analysis is helpful, but it is only shown for two datasets. The authors should clarify why REMC and AZT1D were selected and whether similar interpretability patterns hold across other datasets. The discussion should include a stronger limitations section. Important limitations include greedy search, local optima, risk of overfitting, high computational cost, limited baseline comparison, and reduced generalization in cross-subject settings. Several writing and formatting issues need correction, including typographical errors such as "anaylsis," inconsistent capitalization of dataset names, and overly long paragraphs

in the Discussion. The manuscript would benefit from a clearer contribution statement at the end of the Introduction, listing the main methodological and empirical contributions. The authors should avoid overstating clinical applicability. The datasets are benchmark datasets, and no prospective clinical validation or expert user study is presented. The manuscript should clarify whether SublimeX is intended mainly for prediction, hypothesis generation, or clinical decision support, because the validation requirements differ for each use case. The manuscript is promising and relevant to Pattern Recognition, but it requires major revision to improve methodological rigor, reproducibility, baseline comparison, and clarity of claims.

Our point-by-point responses

Reviewer comment

1. Novelty vs. interval / shapelet / ROCKET / supervised extractors.

The manuscript presents an interesting interpretable feature extraction framework, but the novelty of SublimeX should be stated more clearly. The authors should explain how it differs substantially from existing interval-based, shapelet-based, ROCKET/MiniROCKET, and supervised feature extraction methods.

Our response

We now state the novelty in the Introduction and the new Sec. 1.2 (Related Work), contrasting SublimeX with fixed libraries, shapelet methods, and deep models (Table 1).

- Introduction: "Unlike tsfresh-style fixed libraries, SublimeX searches which regions matter for each task."
- Sec. 1.2: shapelet vs. segment-mean contrast; "combines traceable segment means with supervised search"; Table 1 compares type, # features, multivariate handling (joint vs. per-channel), and interpretability (High/Medium/Low) across fixed libraries, interval/shapelet methods, ROCKET/MiniRocket, and deep learning baselines.
- Methods, Baselines: tsfresh, catch22, MiniRocket, RDST, and 1D CNN under matched protocols.

Reviewer comment

2. Title "time series and spatial data".

The title mentions both "time series and spatial data," but most of the paper focuses on time-series biomedical datasets. The spatial data contribution is not sufficiently demonstrated. The authors should either strengthen the spatial-data evaluation or revise the title and scope.

Our response

We agree that six of seven benchmarks are temporal and that REMC is our sole spatially indexed evaluation (1D epigenomic profiles along genomic position, not 2D imaging). We kept the title and state in Methods, Overview that segment search is agnostic to temporal vs. spatial indexing; REMC is evaluated per cell line (56 models, pooled reporting) with full SHAP interpretation on E003.

- Methods, Overview: "The search is agnostic to whether the index axis is temporal or spatial: it operates on multivariate 1D sequences, and only the domain interpretation of channels and segments differs."
- Methods, Datasets: REMC as "spatial epigenomic gene-expression prediction ... (56 cell lines)";

5-fold CV per cell line, scores pooled across cell lines×folds.
- Results, Domain-Specific Feature Interpretation: REMC labeled "spatial epigenomic profiles"; SHAP on cell line E003.
- Discussion, Limitations: scope limited to 1D genomic locus profiles; no 2D spatial benchmark.

Reviewer comment

3. Abstract competitiveness.

The abstract is informative, but it overstates competitiveness. SublimeX is not the best-performing method on several datasets, including MITBIH, Emotions, MIMIC-IV, PAMAP2, and SVD. The abstract should more cautiously describe the results.

Our response

We toned down competitiveness claims throughout. Overstated wording on AZT1D RMSE, dataset leadership, and state-of-the-art performance was replaced with cautious language.
- Abstract, Results (revised): "ties the best REMC AUC (0.931±0.023)... ranks second on AZT1D (RMSE 30.98 ± 5.65) and MIMIC-IV (AUC 0.818 ± 0.024)"; "uses ∼6–20 descriptors versus thousands for library and kernel methods."
- Results, Performance Comparison (revised): "MiniRocket achieves the lowest mean test RMSE (30.76±5.38); SublimeX is second (30.98±5.65)" with "p=0.39".
- Conclusion (revised): "Interpretability need not come at the cost of performance when feature search is supervised, validation-driven, and constrained to a compact representation" (generalized; no per-dataset score recap).

Reviewer comment

4. Stopping criterion and metric direction.

The methodology needs more detail on the stopping criterion. The paper states that the loop stops when no candidate improves performance, but the exact definition of improvement, tolerance threshold, and validation metric direction should be clearly specified.

Our response

We now define the stopping rule precisely, including strict improvement and metric direction.
- Methods, Overview (revised): "stops once no new segment yields a strict improvement."
- Methods, Iterative Feature Discovery (revised): "only if it improves the running-best validation score"; "When no candidate yields a strict improvement, the loop terminates."
- Methods, Implementation Details: "300 trials per proposed feature"; maximizes accuracy or AUC, minimizes RMSE on AZT1D.

Reviewer comment

5. Fixed 50/50 inner split.

The use of a single deterministic 50/50 train/validation split during feature optimization may introduce instability and overfitting. The authors should justify this choice and discuss whether nested cross-validation or repeated validation would provide more reliable feature selection.

Our response

We added a *Resampling* ablation (Table 4) that redraws the fit/selection partition on every Optuna trial. Resampling was generally slightly worse than the fixed-split baseline across benchmarks; for example, AZT1D RMSE rose from 30.98 ± 5.65 to 31.17 ± 5.85 ($p<0.05$), and PAMAP2 LOSO accuracy fell from 0.860 ± 0.081 to 0.843 ± 0.094. We keep the fixed 50/50 split because it stabilizes candidate scores across trials; Limitations discusses overfitting risk, and nested validation is future work.

- Methods, Evaluation: PAMAP2 LOSO; once-assigned 50/50 fit/selection split on training subjects; anti-leakage and AZT1D per-patient protocol.
- Methods, Iterative Feature Discovery: inner 50/50 split only; refit before test scoring.
- Ablation Study (Table 4), *Resampling*: generally slightly below baseline; e.g. PAMAP2 0.860±0.081 vs. 0.843±0.094; AZT1D RMSE 30.98±5.65 vs. 31.17±5.85.
- Discussion, Limitations: fixed split preferred over Resampling; overfitting to optimization split, especially PAMAP2.
- Discussion, Future Directions: "nested or repeated internal validation."

Reviewer comment

6. PAMAP2 internal vs. LOSO validation.

The internal feature-search validation split does not match the external evaluation design for some datasets, especially PAMAP2 with leave-one-subject-out validation. This likely explains the large validation-test gap. The authors should consider using subject-independent validation during feature search.

Our response

We explain the large PAMAP2 validation–test gap (∼26%): the fit and selection halves share the same training subjects, whereas LOSO tests an entirely unseen subject; the held-out subject never enters the search. **We did not implement** subject-independent validation during every search trial, owing to compute and the 35-page scope; the *Resampling* ablation confirms that redrawn inner splits do not help and hurt PAMAP2 under LOSO.

- Methods, Evaluation: PAMAP2 LOSO; once-assigned fit/selection split on training subjects; held-out subject never enters search.
- Methods, Evaluation / Results: inner split not subject-independent; validation–test gap reported in Results; interpretation in Discussion.
- Results, Incremental Feature Analysis: PAMAP2 gap ∼26% (numbers only).
- Discussion, Summary of Main Findings: PAMAP2 fit/selection vs. LOSO interpretation.
- Table 4: eight-fold PAMAP2 LOSO ablations with mean±SD; *Resampling* underperforms baseline.
- Discussion, Limitations: fixed inner split preferred over *Resampling*; subject-holdout validation during every search trial omitted for compute cost.

Reviewer comment

7. Leakage / outer-fold-only search.

The manuscript should clarify whether feature extraction is performed independently within each training fold only. If any information from validation or test folds influences feature discovery, this

would create data leakage.

Our response

Methods now state explicitly that feature discovery uses the training partition only. Test folds, LOSO subjects, and AZT1D temporal test windows never enter segment search.

- Methods, Evaluation (new): "never enters segment search"; "no leakage from test data into feature discovery"; 5-fold CV, PAMAP2 LOSO, AZT1D temporal split.
- Methods, Iterative Feature Discovery: fit/selection split only (cross-reference to Evaluation).

Reviewer comment

8. HIVE-COTE 2.0, InceptionTime, LSTM/GRU.

The comparison with baselines is incomplete. Important state-of-the-art time series methods such as HIVE-COTEv2, InceptionTime, and LSTM/GRU-based models are only mentioned as limitations but not evaluated. At minimum, the authors should justify why these were excluded.

Our response

We justify the baseline scope in Methods and Limitations: our focus is interpretable, segment-traceable features under matched biomedical splits, so we compare against five complementary baselines (including a 1D CNN on the same folds) and explain why HIVE-COTE, InceptionTime, LSTM, and the UCR archive were excluded.

- Methods, Baselines: "interpretable, segment-traceable features"; hybrid/deep/recurrent methods cited as not directly comparable; 1D CNN on same folds.
- Discussion, Limitations: HIVE-COTEv2, InceptionTime, LSTM "not included"; UCR archive "do not evaluate on."

Reviewer comment

9. Statistical significance.

The authors should provide statistical significance testing across methods. Many reported differences are very small, such as AZT1D RMSE 31.11 vs. 31.18 and REMC AUC 0.931 vs. 0.931, so claims of superior performance require statistical support.

Our response

We added paired Wilcoxon tests and re-ran AZT1D reporting. Table 3 now gives mean±SD of per-patient *test* RMSE rather than a single pooled summary. MiniRocket is best (30.76 ± 5.38); SublimeX is second (30.98 ± 5.65, p=0.39 vs. MiniRocket).

- Methods, Evaluation: "paired two-sided Wilcoxon tests (α=0.05)"; "AZT1D per-patient test RMSE."
- Table 3 (revised): asterisks vs. dataset-best method; AZT1D reported as mean±SD across 23 patients.
- Results, Performance Comparison (revised): "does not differ significantly from MiniRocket (p=0.39)."

Reviewer comment

10. Balance performance vs. interpretability.

The results show that SublimeX is competitive but not consistently superior. The discussion should better balance performance and interpretability rather than implying broad performance advantage.

Our response

The Discussion now balances performance against interpretability, using measured language (MiniRocket best on AZT1D, SublimeX second without a significant gap) and framing interpretability as segment traceability on public benchmarks rather than a performance advantage.

- Discussion, Summary of Main Findings: opens "In this study we presented ... SublimeX"; "competitive" / "stays competitive" wording; MiniRocket best on AZT1D, SublimeX second; baselines lead on several other tasks; PAMAP2 LOSO gap interpreted here (not in Incremental Feature Analysis).
- Abstract, Conclusions: "Interpretability need not sacrifice performance" with traceable channel/transform/segment descriptors.
- Discussion, Comparison to Related Work: predictions related to measurable signal structure; segment descriptors for inspection/SHAP (no deployment or intended-use claim).
- Discussion, Limitations: performance–interpretability trade-offs, baseline scope, cross-subject generalization (PAMAP2).

Reviewer comment

11. Multi-fold ablations.

The ablation study is based on a single fold only, which limits the reliability of the conclusions. The authors should either repeat ablations across multiple folds or clearly tone down the strength of the claims.

Our response

We extended ablations from a single fold to all folds, as reviewers requested. The revision reports mean±SD with Wilcoxon markers in Table 4. REMC ablations use cell line E003 only and are not comparable to the pooled 56-line SublimeX score in Table 3.

- Ablation Study (revised): "report mean±standard deviation across folds"; REMC E003 only; Wilcoxon asterisks in Table 4; *Resampling* variant added.
- Table 4: multi-fold scores, feature counts, and runtimes for eight variants (including *Resampling*).

Reviewer comment

12. Runtime vs. all baselines.

Runtime comparisons need clearer reporting. The manuscript reports runtimes for ablation variants, but does not compare computational cost against all baseline methods under the same hardware and settings.

Our response

Table 3 now reports end-to-end wall-clock time (feature extraction plus training) for SublimeX and all five baselines, measured on the single workstation stated in Implementation Details.

- Table 3: wall-clock time row for SublimeX and all five baselines.

- Methods, Implementation Details: "Experiments were run on a MacBook Pro M2 (12 cores, 32 GB RAM)."
- Results, Performance Comparison: "SublimeX is the slowest method" on large datasets; modest runtime on shorter series.
- Discussion, Limitations: slower than several baselines despite fewer features.

Reviewer comment

13–14. Graphical abstract and Figure 1.

The graphical abstract and Figure 1 contain small text and may be difficult to read. The authors should improve figure resolution, font size, and visual clarity.

Our response

We redrew Figure 1 with larger type and a simplified loop.

- Figure 1: full-width methodology flowchart with enlarged lettering.
- Methods, Overview: "stops once no new segment yields a strict improvement."

Reviewer comment

15. SHAP figure size.

Some figures, especially SHAP visualizations, are too small to interpret in the manuscript. Larger panels or supplementary high-resolution versions should be provided.

Our response

We enlarged Figure 4 to a full-width, six-panel SHAP layout for REMC and AZT1D so panels are legible at print scale.

- Results, Domain-Specific Feature Interpretation: "full six-panel SHAP layouts for REMC (spatial epigenomic profiles) and AZT1D (glucose forecasting) with larger font sizes and only top five features for SHAP beeswarm plot"

Reviewer comment

16. Hyperparameters.

The paper should provide the exact hyperparameters used for LightGBM, CNN, MiniRocket, RDST, tsfresh, and catch22 to ensure reproducibility.

Our response

We now list all hyperparameters explicitly: LightGBM (depth 5, shared readout) in Implementation Details; tsfresh, catch22, MiniRocket, RDST, and CNN settings in Methods, Baselines.

- Methods, Baselines: tsfresh, catch22, MiniRocket, RDST, and CNN settings in prose (replacing "default hyperparameters" and the former supplementary hyperparameter table).

Reviewer comment

17. Open source and reproducibility.

The authors state that SublimeX is open source, but the manuscript should include a permanent repository link, version number, and ideally a reproducibility package for all experiments.

Our response

We now state version, installation, and repository URLs explicitly in Methods. The linked manuscript repository documents end-to-end reproduction of benchmarks, ablations, tables, and figures. Dataset preprocessing is in Supplementary Data.

- Methods, Implementation Details (revised): "published as an open-source Python package on PyPI (version 0.1.2)"; installable via pip; manuscript GitHub URL for reproduction. We ensure the permannence of the link and maintain the Python package as an actively developed package.
- Manuscript repository `https://github.com/Prgrmmrjns/SublimeX-manuscript`: instructions to regenerate tables/figures from saved features or rerun the full experiment pipeline. Note the repo does not contain any datasets, these have to be downloaded beforehand. We provide instructions in the repo and in the Supplementary Data for downloading and reproducing the data as well as the corresponding preprocessed files.

Reviewer comment

18. Definition of "interpretable".

The manuscript should better define what "interpretable" means in this context. Traceability to a signal segment is useful, but interpretability also depends on clinical or biological validity, stability, and expert confirmation.

Our response

We now define interpretability operationally in the manuscript: a feature is interpretable when it is *traceable*—each descriptor names its channel, transform, segment, and aggregation—and when that choice is *stable across folds* and *plausible relative to known biology* where benchmarks allow. We do not claim formal expert confirmation or clinical readiness; SHAP on REMC and AZT1D is our check that ranked segments align with established domain mechanisms.

- Introduction: compact features with "explicit descriptions that experts can inspect directly"; objective (iii) assesses interpretability via post-hoc analyses.
- Results, Transform Composition: fold-stable transform mix and first-feature segment positions (Figure 2).
- Results, Domain-Specific Feature Interpretation: "Each SublimeX feature is readable as a named channel, transform, segment, and aggregation, and we check interpretability by asking whether those choices stay stable across folds and line up with SHAP rankings and known biology where benchmarks allow"; REMC epigenomic hierarchy; AZT1D mean reversion then insulin/carbohydrate inputs (Figure 4).
- Discussion: "traceable, and thereby interpretable"; SHAP recovers meaningful hierarchies "without any domain guidance."

Reviewer comment

19. SHAP on two datasets only.

The SHAP analysis is helpful, but it is only shown for two datasets. The authors should clarify why REMC and AZT1D were selected and whether similar interpretability patterns hold across other datasets.

Our response

We chose REMC and AZT1D for full SHAP figures because they have the clearest domain ground truth and represent the two modalities in the title. For the remaining benchmarks we provide transform-stability (Figure 2) and incremental-feature analyses (Figure 3). There was no space to add SHAP on the remaining datasets. We hope that with our in-depth walkthrough for these two datasets the reader can appreciate the advantages of segment-based traceable features in combination with SHAP analysis for domain interpretation.

- Results, Domain-Specific Feature Interpretation: expanded REMC and AZT1D subsections (biology/clinical context, panel-by-panel SHAP walkthrough); AZT1D mean reversion narrative.

Reviewer comment

20. Stronger limitations section.

The discussion should include a stronger limitations section. Important limitations include greedy search, local optima, risk of overfitting, high computational cost, limited baseline comparison, and reduced generalization in cross-subject settings.

Our response

The Limitations are now a dedicated, multi-paragraph subsection, with Future Directions kept separate.

- Discussion, Limitations: greedy search; runtime; PAMAP2 gap and *Resampling* ablation; HIVE-COTE/InceptionTime/LSTM; UCR; slower than lighter baselines; benchmark scope (no UCR).
- Discussion, Future Directions: nested validation; learned transforms; efficient search.

Reviewer comment

21. Writing and formatting.

Several writing and formatting issues need correction, including typographical errors such as "anaylsis," inconsistent capitalization of dataset names, and overly long paragraphs in the Discussion.

Our response

We corrected the typographical errors, standardized dataset names, moderated performance claims, and tightened the wording and flow throughout.

- Introduction and Methods: Sec. 1.1 Objectives + Sec. 1.2 Related Work; smoother transitions and reduced repetition.
- Discussion: shorter paragraphs; PAMAP2 interpretation in Summary; Limitations and Future Directions separated.
- Conclusion: three generalized sentences (no dataset-specific score recap).
- Abstract and body: "MIT-BIH" (not "MITBIH").

Reviewer comment

22. Contribution statement.

The manuscript would benefit from a clearer contribution statement at the end of the Introduction, listing the main methodological and empirical contributions.

Our response

We added an explicit contribution statement: Sec. 1.1 (Objectives) lists three numbered aims, followed by Sec. 1.2 (Related Work), both before Methods.

- Sec. 1.1: "The objectives of our study are as follows: (i) ... (ii) ... (iii) ..."
- Sec. 1.2: Related Work and Table 1 (method taxonomy, multivariate handling, interpretability) before Methods.

Reviewer comment

23. Clinical applicability.

The authors should avoid overstating clinical applicability. The datasets are benchmark datasets, and no prospective clinical validation or expert user study is presented. The manuscript should clarify whether SublimeX is intended mainly for prediction, hypothesis generation, or clinical decision support, because the validation requirements differ for each use case.

Our response

We do not assign SublimeX to prediction, hypothesis generation, or clinical decision support, and we do not claim clinical readiness. We state the need for inspectable predictions indirectly: many biomedical tasks require outputs tied to measurable signal structure, and SublimeX yields segment-traceable descriptors (channel, transform, location) that support such inspection on our public benchmarks—illustrated with SHAP and fold stability, not prospective validation.

- Abstract, Introduction (revised): "inform clinical and scientific analysis" (motivation only; not a claim about intended use).
- Discussion, Comparison to Related Work (revised): removed "Biomedical deployment places constraints" and "attractive for biomedical use"; now "Many biomedical modeling tasks require predictions that can be related to measurable signal structure ... SublimeX represents each sample through segment-based features with explicit, human-readable descriptions."
- Discussion, Comparison to Related Work: MIMIC sentence now "segment-traceable features tied to identifiable ICU signal regions" (replacing "support clinical validation").

Reviewer comment

Reviewer #2 — metareview

The manuscript presents an interesting and relevant approach to interpretable feature extraction for time series and spatial data, addressing an important gap between predictive performance and model transparency. The proposed framework is well motivated and demonstrates competitive results across multiple datasets, with a clear emphasis on interpretability. However, several aspects require improvement to strengthen the contribution. In particular, the objectives and novelty of the method should be more explicitly stated, and the distinction from existing approaches clarified. The experimental section would benefit from more rigorous statistical analysis, clearer reporting of reproducibility details, and stronger justification of evaluation protocols. Additionally, some claims

regarding performance and interpretability should be moderated or better supported by quantitative evidence. The manuscript is generally well structured, but certain sections could be streamlined for clarity, and minor language editing is recommended. Overall, the work has merit, but revisions are needed to improve clarity, rigor, and positioning.

Our response

We thank Reviewer 2 for this constructive summary. Every theme is addressed in Q1–Q8 and in the point-by-point responses below, with quoted wording from the revised manuscript. Where a request could not be fully met, we explain the constraint (chiefly the 35-page limit) and what we provided instead.

- See Q1–Q8 and Reviewer 1 items 1–23.

We hope this letter and the revised manuscript address the reviewers' concerns.

Sincerely,
The Authors